Session 4 – Linear classifiers, data-dependent priors and synthesis

Goal. By the end of the session you should be able to implement PBGD, certify a Gaussian posterior exactly and by Monte Carlo, learn a prior on part of the data, and put all the results of the four sessions together.

To reread before the session. Sections 4.2, 4.3, 6.4 and 7.4 of the notes; notebook 5.

Timing (3 hours).

Part 4.1	75 min
Part 4.2	50 min
Part 4.3	30 min
Part 4.4	25 min

To hand in at the end of the session. Your notebook, with the tables of Parts 4.1 and 4.2, the decision table of Part 4.3, and the outline of the final report.

Part 4.1 – (⋆) PAC-Bayesian learning of linear classifiers and Monte Carlo certificates (Sections 4.3 and 7.4)

Consider linear classifiers $h_{\mathbf{v}}(x) = \operatorname{sign}\langle \mathbf{v}, \tilde{x}\rangle$ on the standardized features augmented with a constant 1 ($\tilde{x} = (x, 1)$), the prior $P = \mathcal{N}(\mathbf{0}, I)$ and the posteriors $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I)$.

1. Implement the exact empirical Gibbs risk $\widehat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m}\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle / \|\tilde{x}_i\|)$ and the PBGD objective $C\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle/\|\tilde{x}_i\|) + \frac{1}{2}\|\mathbf{w}\|^2$ with its gradient; minimize it with L-BFGS, starting from a linear SVM.

2. On `australian`, `pima`, `wdbc`, `spambase`, compute Seeger's certificate on the Gibbs risk for 13 values of C in $[10^{-2}, 10^2]$, select the best one, and compare the test risk of the vote $\operatorname{sign}\langle \mathbf{w}, \tilde{x}\rangle$ with a linear SVM ($C = 1$). Plot the certificate as a function of C.

3. Pretend that the Gibbs risk has no closed form: estimate it with $N = 2000$ classifiers drawn from $Q_{\mathbf{w}}$ and compute the Monte Carlo certificate (Section 4.3), with $\delta/2$ for each step. How much is lost?

4. Why can C be selected by minimizing the certificate without any union bound?

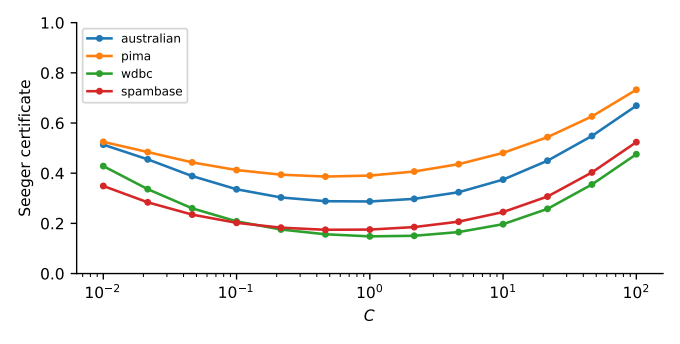
Solution

Algorithm 13: PBGD with certificate-based choice of C, and Monte Carlo certificate

for C *in a geometric grid* **do**
  $\mathbf{w}_C \leftarrow \arg\min_{\mathbf{w}} \; C\sum_i \Phi\big(-\frac{y_i\langle \mathbf{w}, \tilde{x}_i\rangle}{\|\tilde{x}_i\|}\big) + \frac{1}{2}\|\mathbf{w}\|^2$; ▷ *L-BFGS, warm start at the SVM*
  $B_C \leftarrow \overline{\mathrm{kl}}^{-1}\big(\widehat{R}_S(G_{Q_{\mathbf{w}_C}}), \frac{\|\mathbf{w}_C\|^2/2 + \ln(2\sqrt{m}/\delta)}{m}\big)$;
$\mathbf{w} \leftarrow \mathbf{w}_{C^\star}$ with $C^\star = \arg\min_C B_C$;
; ▷ *Monte Carlo version*
draw $\mathbf{v}_1, \dots, \mathbf{v}_N \sim \mathcal{N}(\mathbf{w}, I)$; $\hat{r}_N \leftarrow \frac{1}{Nm}\sum_{j,i} \mathbb{1}\big[h_{\mathbf{v}_j}(x_i) \neq y_i\big]$;
$\bar{r} \leftarrow \overline{\mathrm{kl}}^{-1}\big(\hat{r}_N, \ln(4/\delta)/N\big)$; certificate $\leftarrow \overline{\mathrm{kl}}^{-1}\big(\bar{r}, \frac{\|\mathbf{w}\|^2/2 + \ln(4\sqrt{m}/\delta)}{m}\big)$;

Python: function `probit_risk, pbgd, mc_certificate, partD` of `tp_solutions.py`.

dataset	SVM test	PBGD (best C) $C^\star$	$\|\mathbf{w}\|$	$\widehat{R}_S(G)$	Seeger cert.	MC cert.	test vote	test Gibbs
australian	0.150	1.00	5.68	0.155	0.286	0.318	0.140	0.164
pima	0.229	0.46	4.11	0.266	0.379	0.413	0.242	0.270
wdbc	0.047	1.00	5.06	0.052	0.150	0.176	0.041	0.073
spambase	0.075	0.46	10.10	0.110	0.180	0.206	0.078	0.113



2. The certificate is a U-shaped function of C (figure): for small C the posterior stays close to the prior and its Gibbs risk is near $\frac{1}{2}$; for large C the norm, hence the KL, explodes. The selected $C^\star$ is between 0.5 and 1 (average over the splits). The vote of PBGD is as accurate as the linear SVM (0.140 vs 0.150 on `australian`, 0.041 vs 0.047 on `wdbc`, slightly worse on `pima`), and it comes with a certificate: 0.150 on `wdbc` and 0.180 on `spambase`. The certificate bounds the risk of the Gibbs classifier (test value 0.073 and 0.113); the risk of the vote is at most twice this value, and is actually smaller.
3. With $N = 2000$ samples, the Monte Carlo certificate is only 0.02–0.035 larger (0.176 instead of 0.150 on `wdbc`): the price of not knowing the Gibbs risk exactly is small, which is what makes PAC-Bayesian certificates for neural networks possible.
4. C only indexes posteriors $Q_{\mathbf{w}_C}$, all compared with the same prior in the same bound (Seeger's): the certificate is valid simultaneously for all of them (Theorem 6.2). This is a real advantage over the SVM, whose C must be chosen by cross-validation.

Part 4.2 – (⋆) Data-dependent priors for linear classifiers (Sections 4.2 and 6.4)

1. Split the training set into S_1 (a fraction f) and S_2. Learn a prior centre $\mathbf{w}_0$ on S_1 (PBGD with $C = 1$), then run PBGD on S_2 with the prior $\mathcal{N}(\mathbf{w}_0, I)$, i.e. with the regularizer $\frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$, and compute Seeger's certificate on S_2 only ($\mathrm{KL} = \frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$), selecting C on the grid of Part 4.1.
2. Compare $f \in \{0, 0.2, 0.5\}$ ($f = 0$ is Part 4.1) on the four datasets: size of S_2, KL, certificate, test risk.
3. Which examples may the posterior use? Why is the certificate valid although $\mathbf{w}_0$ depends on the data?

Solution

Algorithm 14: Data-dependent Gaussian prior

split the training set into S_1 (f) and S_2 ($1 - f$), $m_2 = |S_2|$;
$\mathbf{w}_0 \leftarrow$ PBGD on S_1 (prior $\mathcal{N}(0, I)$, $C = 1$) ; ▷ $\mathbf{w}_0 = 0$ *if* $f = 0$
for C *in the grid* **do**
  $\mathbf{w}_C \leftarrow \arg\min_{\mathbf{w}} C \sum_{i \in S_2} \Phi\big(-\frac{y_i \langle \mathbf{w}, \tilde{x}_i \rangle}{\|\tilde{x}_i\|}\big) + \frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$;
  $B_C \leftarrow \overline{\mathrm{kl}}^{-1}\big(\widehat{R}_{S_2}(G_{Q_{\mathbf{w}_C}}), \frac{\|\mathbf{w}_C - \mathbf{w}_0\|^2/2 + \ln(2\sqrt{m_2}/\delta)}{m_2}\big)$;
return the best C and its certificate;

Python: function `partE` of `tp_solutions.py`.

dataset	f	m_2	KL	certificate	test risk (vote)
australian	0.0	483	16.11	0.286	0.140
australian	0.2	387	2.82	0.246	0.132
australian	0.5	242	1.73	0.243	0.142
pima	0.0	537	8.45	0.379	0.242
pima	0.2	430	3.33	0.351	0.235
pima	0.5	269	1.01	0.366	0.242
wdbc	0.0	398	12.81	0.150	0.041
wdbc	0.2	319	4.58	0.119	0.039
wdbc	0.5	199	1.83	0.123	0.041
spambase	0.0	3217	51.03	0.180	0.078
spambase	0.2	2574	12.59	0.139	0.079
spambase	0.5	1609	4.55	0.134	0.078

2. Learning the centre of the prior on a fifth of the data divides the KL by 3 to 6 (16.1 → 2.8 on `australian`, 51.0 → 12.6 on `spambase`) and improves every certificate, although it is computed on fewer examples: 0.286 → 0.246 on `australian`, 0.150 → 0.119 on `wdbc`, 0.180 → 0.139 on `spambase`. Going to $f = 0.5$ helps further on the largest dataset (`spambase`: 0.134) but not on the small ones, where S_2 becomes too small (`pima`: 0.351 → 0.366): the same trade-off as in Figure 28 of the notes. The test risk of the vote is essentially unchanged.
3. In the version implemented here the posterior only uses S_2, but it could use all the data: only the *empirical term* of the bound has to be computed on S_2 (Proposition 6.5). The certificate is valid because, conditionally on S_1, the prior $\mathcal{N}(\mathbf{w}_0, I)$ is fixed and S_2 is an independent i.i.d. sample: the PAC-Bayesian theorem applies conditionally, and integrating over S_1 gives the result. The fraction f must be fixed in advance (or chosen on a grid with a union bound).

Part 4.3 – Synthesis: which certificate for which situation?

Using your results of the four sessions (and not new computations), fill in the following decision table. For each situation, say which guarantee you would report (test-set bound, first-order, tandem or C-bound certificate, Seeger's bound on a Gibbs classifier, with or without data-dependent prior), and justify your choice in one sentence with a number from your experiments.

situation	guarantee to report, and why
a forest has been learned, and 30% of the data can be set aside	
a small dataset (a few hundred examples)	
the weights of the vote are learned from the data	
several models are compared and one is selected	
an imbalanced problem	
a boosted vote of stumps	
a stochastic (Gibbs) linear classifier	

Solution

situation	guarantee to report, and why
fixed forest, 30% of the data available	the *test-set bound* of the vote: 3 to 5 times tighter than any self-bounding certificate (Part 3.2: 0.071 vs 0.274 on `spambase`) for almost the same vote.
small dataset	the *first-order* OOB certificate, estimated on the largest sample (n_1) and multiplied by 2 only (Part 1.3); but expect it to be loose, and prefer a hold-out certificate if the vote matters more than the bound.
weights learned from the data	the *tandem* certificate of the *TND-optimized* posterior: it is valid for a data-dependent posterior and learning with it preserves the vote (Part 2.1); never learn with the first-order bound.
model selection	certificates computed with δ/K and the smallest one, *if they are non-vacuous* (Part 2.4); otherwise cross-validation followed by a hold-out certificate.
imbalanced problem	a certificate on the *balanced error*, class by class with a union bound (Part 3.3); the 0-1 certificate cannot even beat the trivial classifier.
boosted vote of stumps	a *hold-out* certificate of the vote (Part 3.4): the Gibbs risk of weak learners makes every Gibbs-based certificate vacuous.
stochastic linear classifier	*Seeger's bound* on the Gibbs risk (exact probit risk, or Monte Carlo with a second kl inversion), with a *data-dependent prior* on a fifth of the data (Parts 4.1 and 4.2: 0.119 on `wdbc`).

Part 4.4 – Report (to finish after the session)

Summarize your results in a short report (3–4 pages) containing the tables of Parts 1.3 and 2.1, two figures of your choice and a discussion. As in the first practical session, compare methods that are comparable and give standard deviations; remember that a certificate is not an estimate of the risk, and comment on its tightness, not only on its value; finally, report every choice made using the data (grids, splits, selected bounds), since each of them may require a union bound.

Solution

A good report states the four lessons of the session. First, certificates are only informative when the voters are evaluated on enough examples (a few thousand OOB examples for a random forest): on small datasets they are vacuous, and this is not a failure of the implementation. Second, the choice of the bound matters twice: as a certificate, the first-order bound is often the tightest on small and medium datasets and the tandem bound on large ones; as a learning objective, only the second-order bound preserves the diversity and the accuracy of the vote. Third, for a fixed vote the hold-out certificate is much tighter, while self-bounding certificates shine when they are used to learn, to select, or to certify stochastic predictors, especially with data-dependent priors. Fourth, every data-dependent choice has a price, usually small ($\ln K$), but it must be paid.